

Douglas Gordon Daly

Principal AI Architect / AI Platform Engineering Lead

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PROFESSIONAL SUMMARY

Senior technical leader with deep expertise architecting enterprise AI/ML platforms, governed GenAI and tool-calling systems, and operational intelligence solutions. Built governed analytics assistants, monitoring frameworks, and reusable data/AI platforms with strong emphasis on controls, observability, and auditable execution. Experienced across cloud-native AI infrastructure, large-scale data platforms, executive alignment, and technical mentorship. Known for translating ambiguous business needs into scalable, reliable analytical systems that improve decision-making and operational performance.

CORE EXPERTISE

- AI Platform Architecture - enterprise AI platforms, governed analytics assistants, operational intelligence systems, reusable AI/analytics frameworks
- GenAI & Tool-Calling Systems - LLM applications, structured outputs, tool orchestration, RAG/GraphRAG systems, prompt engineering, evaluation, traceability
- Governance & Operational Controls - responsible AI adoption, guardrails, auditable execution, analyst-in-the-loop workflows, policy-aware system design
- Observability & Reliability - monitoring frameworks, incident detection, alerting, operational health, SLIs/SLOs, SEV analysis, on-call-friendly design
- Cloud & Data Platforms - Python, SQL, BigQuery, AWS SageMaker/S3/Terraform, GCP, Docker, data profiling, validation, cross-system integration
- Technical Leadership - architecture direction, executive alignment, technical roadmaps, stakeholder translation, mentoring, developer enablement

PROFESSIONAL EXPERIENCE

Principal Consultant | Daly Engineers LLC | Jan 2026 - Present

Consolidated Edison (via ABC Consulting)

- Architected and delivered a governed analytics assistant leveraging Gemini, BigQuery, and tool-calling architectures, embedding analyst rules, operational controls, and auditable execution for AI-enabled construction and financial analytics.
- Developed automated frameworks for profiling, type inference, duplicate detection, candidate key analysis, and cross-system join validation across 100+ source tables and millions of operational and financial records, enhancing data quality and interoperability.
- Defined a phased analytical maturity roadmap for anomaly detection, construction analytics, and AI-enabled decision support, guiding strategic adoption of advanced analytics capabilities.

NBCUniversal (via Apex Systems)

- Architected AWS-based AI and graph analytics pipelines leveraging SageMaker, Terraform, S3, and containerized workloads to enable scalable, cloud-native analytics solutions.
- Evaluated and demonstrated emerging graph analytics and graph machine learning technologies—including Neptune, GraphStorm, Bedrock, and LangGraph—to support dependency analysis and operational intelligence initiatives.

AI Instructor | SimpliLearn & Interview Kickstart | March 2025 - Present

- Developed and delivered practical GenAI course materials, demos, and exercises on chatbots, RAG, agentic workflows, multimodal AI, diffusion models, evaluation, and enterprise AI governance when existing curriculum required deeper technical examples and clearer hands-on instruction.
- Led multi-month AI and machine learning training programs for engineers and data scientists, mentoring learners on AI architecture, implementation tradeoffs, governance, and production-oriented GenAI development workflows.

Data Scientist | Intuit (via The Cydio Group) | Apr 2025 - Jul 2025

- Consolidated fragmented reporting into unified Tableau dashboards and automated executive reporting workflows using Python and SQL, significantly reducing report preparation time and driving operational efficiency.
- Collaborated with finance, marketing, and product stakeholders to translate attribution and performance data into actionable business recommendations, strengthening cross-functional alignment.

Data Scientist | Meta | October 2021 - January 2025

- Designed and developed the ReMon framework, empowering engineering teams to implement comprehensive monitoring and alerting systems without deep internal tooling expertise; adopted across Meta FinTech and enabled automatic detection of over 15 SEVs in 2024.
- Built forecasting and performance-management frameworks that translated engineering initiatives into projected reliability outcomes, directly informing planning, goal setting, and leadership decision-making across business pillars.

- Delivered impact sizing, ROI, and risk exposure analyses to guide investment and prioritization decisions for platform reliability, privacy compliance, authentication, testing infrastructure, and data platform initiatives.
- Built attribution frameworks linking incidents, alerts, defects, and operational metrics to engineering organizations and service ownership structures, enabling more accurate accountability and performance measurement.

Data Scientist | Samsung Electronics (via Harvey Nash) | Dec 2020 - Oct 2021

- Developed and deployed gradient-boosting propensity models to optimize digital marketing campaigns, driving a 146% increase in conversions and a 229% rise in campaign-attributed revenue while automating audience segmentation and targeting workflows for improved efficiency and scalability.

Head of Data Science | Facteus | Oct 2017 - Nov 2020

- Directed data science initiatives leveraging a 30B+ transaction dataset to develop analytical products and market intelligence capabilities through large-scale feature engineering, merchant intelligence, and consumer-spending analytics.

Expert SEO Solutions Architect | Nike (via Intersoft Inc.) | Nov 2016 - Sep 2017

- Architected a scalable SEO and information-architecture framework for Nike's product discovery, aligning customer navigation with search engine indexation to support a rapidly growing product catalog.

Senior Manager, Operations Analysis | Capital One | Jan 2012 - Dec 2016

- Conceived, architected, and led adoption of RITHM, a real-time operational intelligence platform aggregating monitoring signals by business capability and system ownership, reducing major incident detection times from hours to minutes.
- Secured executive sponsorship and partnered with CTO-level leadership to define enterprise monitoring strategy, expanding Splunk-based prototypes into a scalable Kafka and Elastic Stack solution aligned with open-source modernization initiatives.
- Developed predictive capacity-management models for customer-facing banking platforms, enabling infrastructure planning and risk management through demand forecasting based on business-process and temporal drivers.

Senior Principal Systems Engineer / Program Manager | Raytheon | Mar 2005 - Dec 2011

- Led systems engineering organizations of 40-50 engineers specializing in modeling, simulation, analysis, and advanced aerospace systems, providing technical leadership, career development, and proposal support for programs valued up to \$50M.
- Turned around an over-budget classified fixed-price technology program by establishing execution plans, aligning engineering teams, and proactively managing technical and program risks; delivered under budget and nearly doubled the original profit target on a historically challenging delivery effort.

SELECTED AI PROJECTS

GraphRAG Engine & LoRA Coverage Assistant | GitHub | 2025

- Built a GraphRAG system for multi-hop benefits reasoning with fully traceable retrieval paths across entities, clauses, and supporting evidence.
- Fine-tuned a LoRA adapter on a 7B open-source LLM to enforce a strict JSON schema (decision/reason/disclaimer) for reliable downstream automation.

Agent-to-Agent Protocol MVPs | GitHub | 2026

- Agent Olympics: built a small evaluation runner that benchmarks a generalist agent vs discovered specialists across three task categories with explicit scoring criteria and summary reporting.
- AI PEMDAS: built an instrumented coordination workflow where an evaluator routes sub-steps to ADD vs MULT agents and logs calls for end-to-end traceability (FastAPI JSON-RPC).

AI-Assisted Resume Generation System | GitHub | 2026

- Built an evidence-driven resume generation pipeline that converts a master resume into canonical JSON, extracts candidate evidence, maps evidence to market-derived role signals, and assembles targeted resume variants through structured LLM workflows.
- Implemented versioned artifacts, human review checkpoints, traceable evidence IDs, and DOCX/PDF rendering controls to separate source evidence, content selection, positioning, and presentation layout.

EDUCATION

MBA | UCLA Anderson School of Management | Top Honors | GPA: 3.9

M.S., Applied Statistics | Rochester Institute of Technology | GPA: 3.9

M.S., Electrical Engineering | University of Washington | GPA: 3.7

B.S., Physics (Minor: Mathematics) | Humboldt State University | GPA: 3.4

PATENTS & RECOGNITION

- Lead Inventor, U.S. Patent #8635622: Method and System for Resource Management Using Fuzzy Logic Timeline Filling (2014)
- Capital One Business Technology Innovation Award (2014)
- Conference Chair & Speaker - Customer Analytics Innovation Summit (2016)